Social Networks and their Analysis

INFLUENCE MODELING

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Social Networks and their Analysis

Basic Concepts

- *Network Representation*
- *Tie Strength*
- *Key Players*
- *Cohesion*

Influence Modeling

- **Linear Threshold Model**
- Independent Cascade Model
- Influence vs. Correlation

Common properties of influence modeling methods

- A social network is represented as a *directed graph, with each actor being one node;*
- Each node is started as active or inactive;
- A node, once activated, will activate its neighboring nodes;
- Once a node is activated, this node cannot be deactivated.

Linear Threshold Model

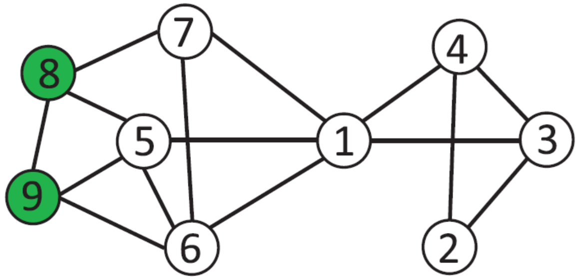
An actor (~node) would take an action if the number of his friends who have taken the action exceeds a certain threshold

- Each node v *chooses a threshold* θ_v *randomly from a uniform* distribution in an interval between 0 and 1.
- In each discrete step, all nodes that were active in the previous step remain active
- The nodes satisfying

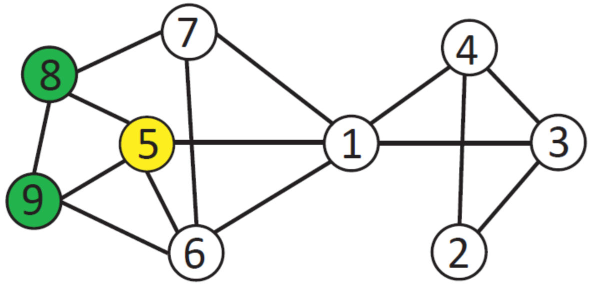
$$\sum_{w \in N(v),\, w \text{ is active}} b_{w,v} \geq \theta_v$$

will also be activated.

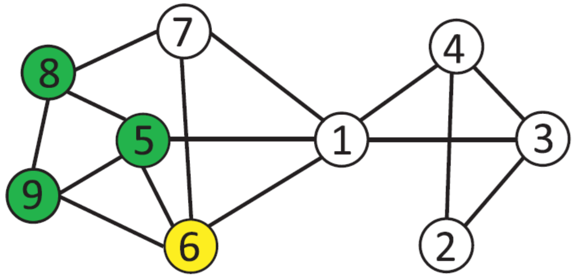
Linear Threshold Model- Diffusion Process
(Threshold = 50%)



Step 0



Step 1



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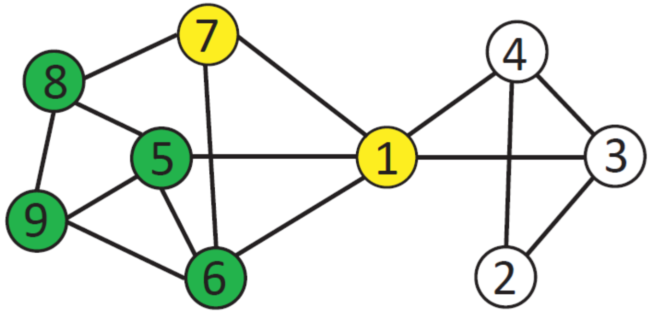

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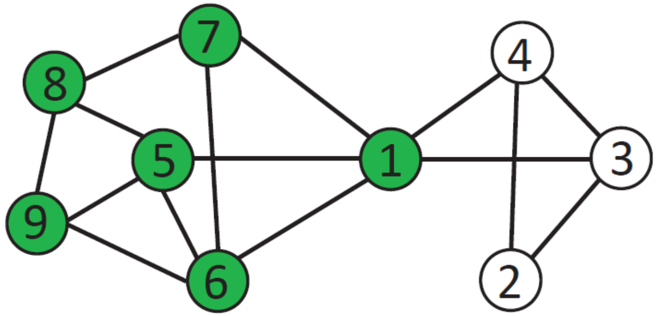

Final Stage

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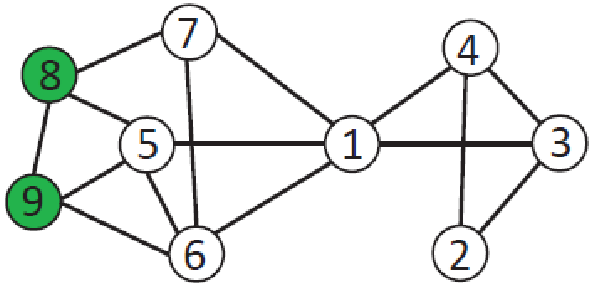
Independent Cascade Model (ICM)

~ focuses on the senders rather than the receivers

- A node w, *once activated* at step t, *has one chance to activate each of its neighbors* ***randomly***
  - For a neighboring node (say, v), *the activation succeeds* with probability $p_{w,v}$ (e.g., $p = 0.5$)
- If the activation succeeds, then v *will become active at step* $t + 1$
- In the subsequent rounds, w *will not attempt to activate* v *anymore.*
- *The diffusion process* starts with an initial activated set of nodes and then continues until no further activation is possible

Independent Cascade Model- Diffusion Process



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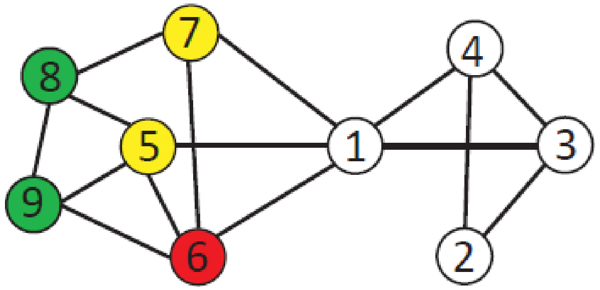

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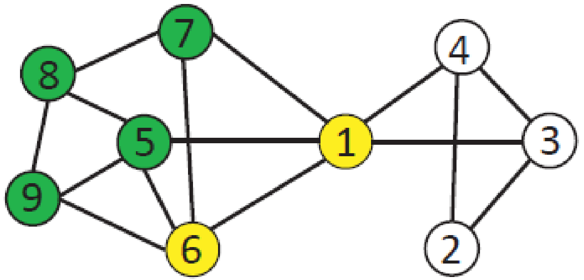

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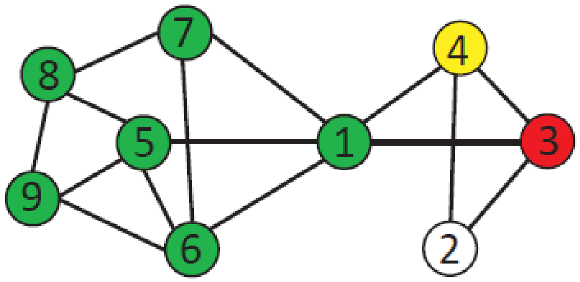

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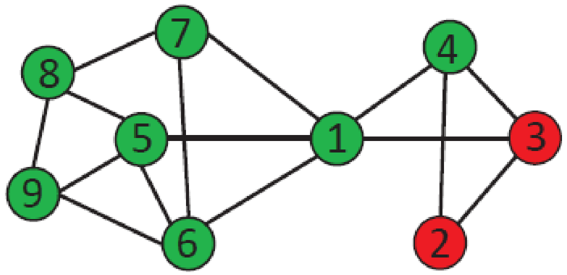
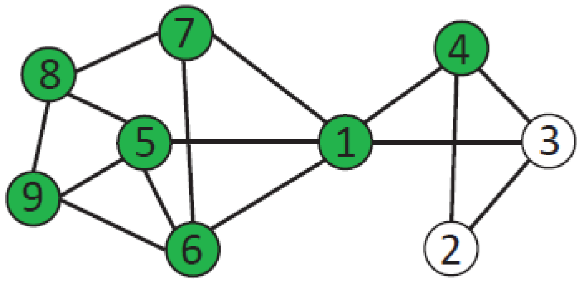


Step 4



Final Stage

Influence maximization

- Given a network and a parameter k*, which k nodes should be selected to be in the* activation set B *in order to maximize the influence in terms of active nodes at the* end?
- Let $\sigma(B)$ *denote the expected number of nodes that can be influenced by* B*, the* optimization problem can be formulated as follows: $\max_{B \subseteq V \& |B| \leq k} \sigma(B)$

Influence maximization – a greedy approach

~ **an NP-hard problem**

x but it can be proved that the greedy approach yields solutions that are 63 % of the optimal.

- **Greedy approach:**
  - ***Start with*** $B = \emptyset$
  - ***Evaluate*** $\sigma(v)$ ***for each node, and pick the node with maximum*** $\sigma(v)$ ***as the first node*** v_1 ***to form*** $B = \{v_1\}$
  - ***Select a node that will increase*** $\sigma(B)$ ***most if the node is included in*** B***.***

Essentially, we greedily find a node $v \in V \backslash B$ **such that** $\underset{v \in V \backslash B}{\operatorname{argmax}} \sigma(B \cup \{v\})$

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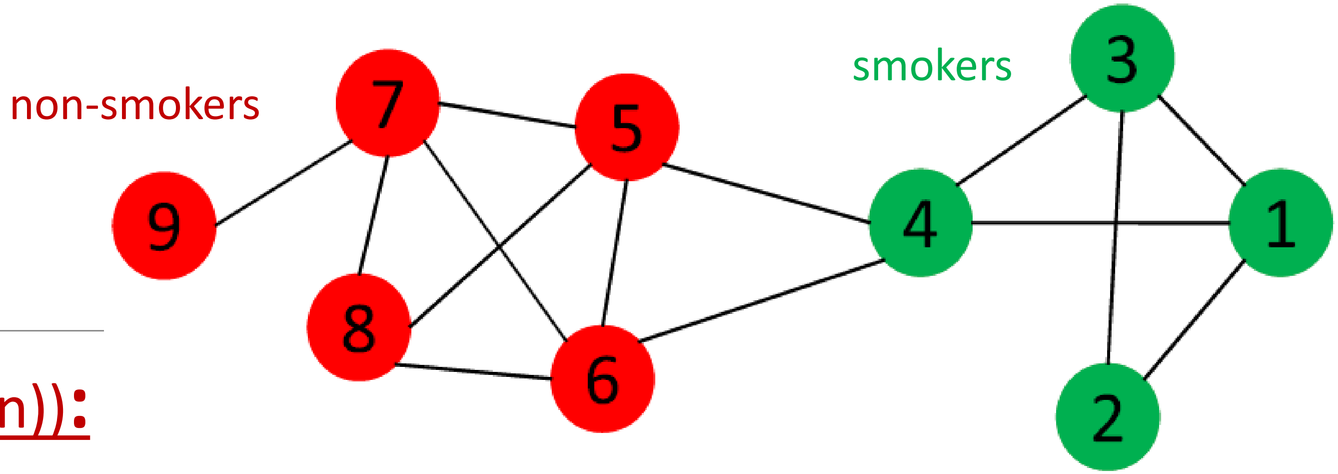
Influence vs. correlation in social networks

User attributes and behaviors quite often tend to correlate with their social networks:

- suppose a binary attribute is associated with each node
  - e.g., whether or not being a smoker
- if the attribute is correlated with the network, the actors are expected to share the same attribute values that would be positively correlated with social connections
- i.e., smokers are more likely to interact with other smokers, and non-smokers with non-smokers

➔ **Test for correlation:** If the fraction of edges linking nodes with different attribute values occur significantly less often than the expected probability, then there is evidence of correlations.

Test for correlation



EXAMPLE (evidence of correlation))**:**

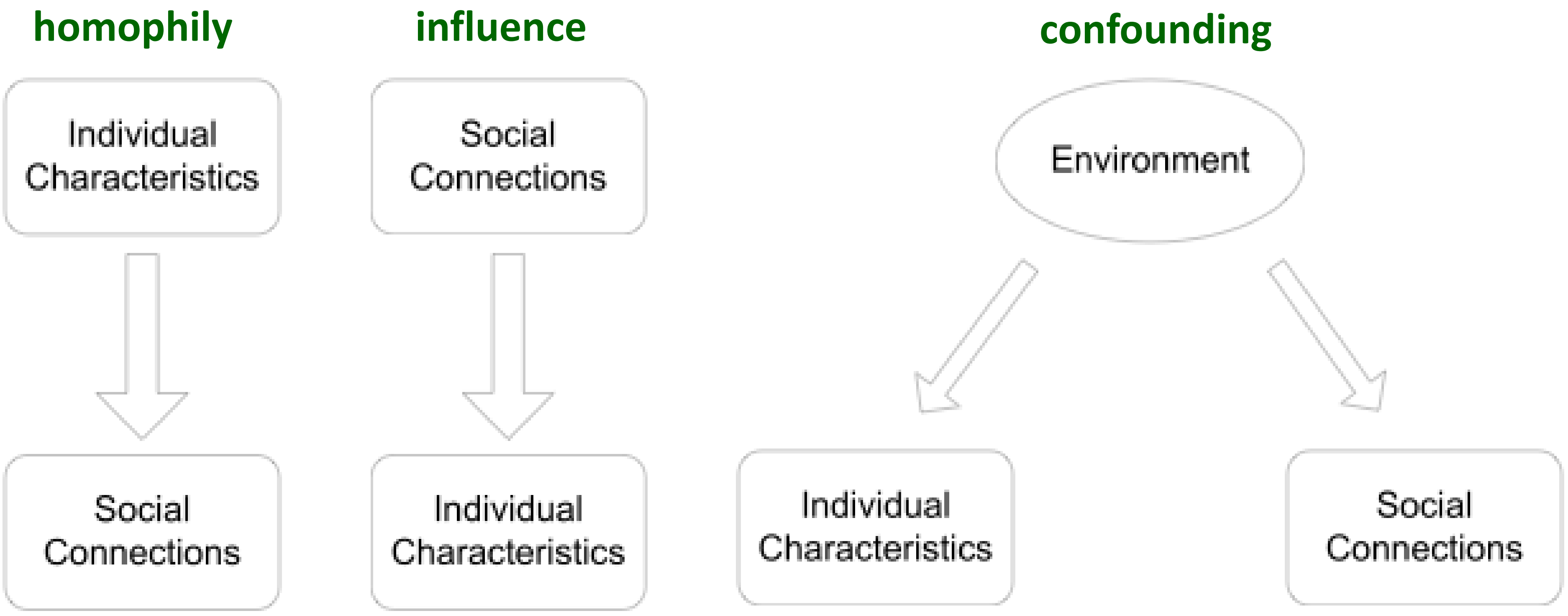
- if connections are independent of the smoking behavior: p fraction are smokers (4/9); $1 - p$ non-smokers (5/9)
- one edge is expected to connect two smokers with probability $p \times p$,
- *two non-smokers with* probability: $(1 - p) \times (1 - p)$
- *a smoker and a non-smoker:* $2\,p\,(1 - p)$
- In the above graph, red nodes denote non-smokers, and green ones are smokers. If there is no correlation, then the probability of one edge connecting a smoker and a non-smoker is 2 × *4/9* × *5/9 = 49%.*
- *But here, the fraction is 2/14 = 14% < 49%, so* this network demonstrates some degree of correlation with respect to smoking behavior.

Correlation in social networks

- in real social networks, there exist correlations between the behavior or attributes of adjacent actors / nodes
- social processes explaining correlation:
  - **Homophily:** the tendency to link to others who share certain similarity with us
    ´birds of a feather flock together´
  - **Confounding:** correlation between actors / nodes can also be forged due to external influences from the environment
    ´individuals living in the same city are more likely to become friends than random individuals´
  - **Influence:** process **causing** behavioral correlations between adjacent actors
    ´if most of one's friends switch to a mobile company, he might be influenced by his friends and switch to the company as well.´

Correlation in social networks

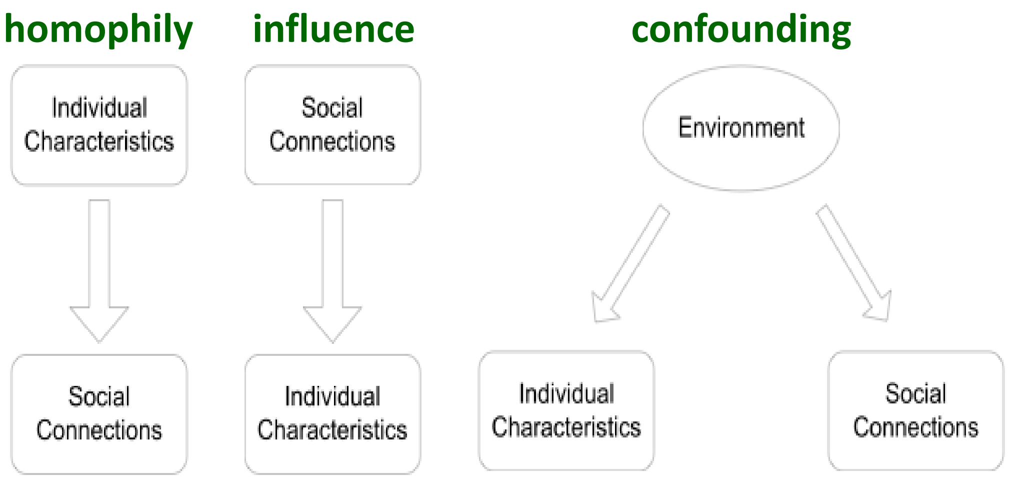
Social processes explaining correlation:



Correlation in social networks

Why distinguish influence from homophily and confounding?

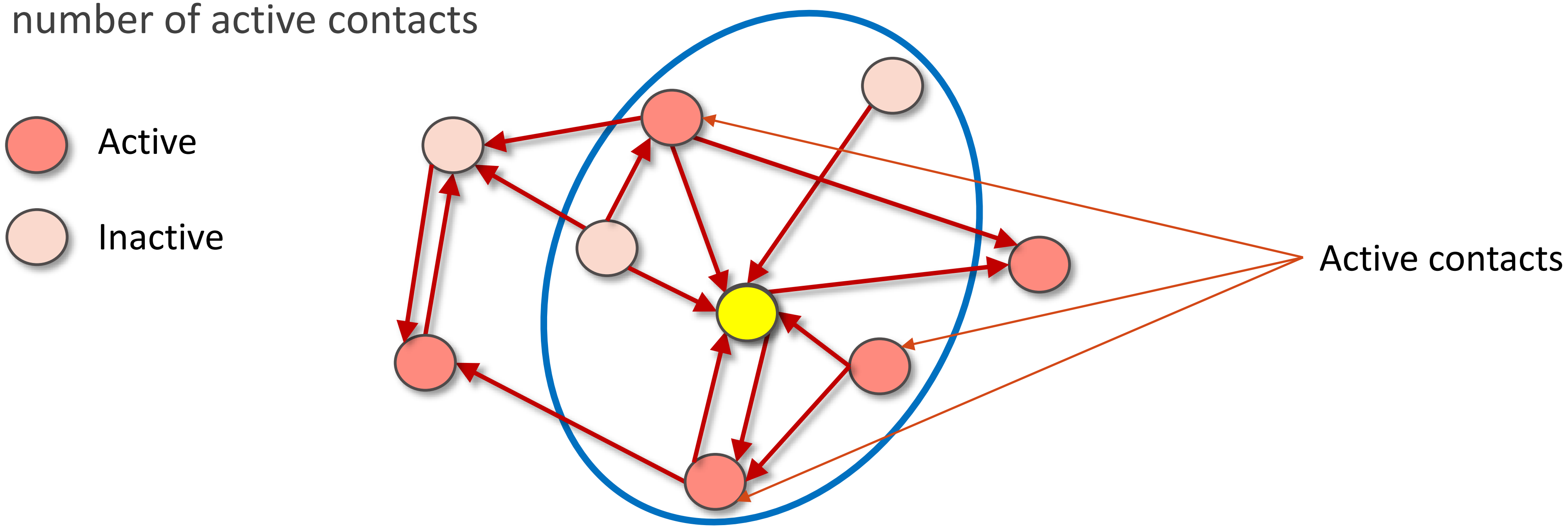
- **Analysis:** to predict the dynamics of the system – whether a new norm of behavior, technology, or idea can diffuse like an epidemic
- **Design:** to design a system for inducing a particular behavior, e.g.
  - Vaccination strategies (random, targeting a demographic group, random acquaintances, etc.)
  - Viral marketing campaigns



Influence vs. Correlation

In many studies about influence modeling, influence is determined by timestamps

- An edge (u, v): the node u can influence the node v at discrete time steps: $t = 0, 1, 2, \ldots$
- For each t, every inactive node becomes active with the probability $p(a)$, where a is the number of active contacts



Influence vs. Correlation

The probability of one node being active is a logistic function of the number of his active friends as follows

$$p(a) = \frac{e^{\alpha \ln(a+1)+\beta}}{1 + e^{\alpha \ln(a+1)+\beta}} \quad \text{equivalently} \quad \ln\left(\frac{p(a)}{1 - p(a)}\right) = \alpha \ln(a+1) + \beta$$

- *a is the number of active friends,*
- *α is the social correlation coefficient*
- *β is a constant to* explain the innate bias for activation
- $\ln(a+1)$ as an explanatory variable

Activation likelihood

Suppose at a time point t, $Y_{a,t}$ users with a active friends become active, and $N_{a,t}$ users who also have a active friends yet stay inactive at time t.

The **likelihood of activation at time t** is
$$\prod_a p(a)^{Y_{a,t}}\left(1-p(a)\right)^{N_{a,t}}$$

The likelihood of the whole sequence $\{Y_{a,t}, N_{a,t}\}_{t=0}^{T}$ is

$Y_{a,t}$ and $N_{a,t}$ decreases quickly with a, only $a \leq R$ is used for some constant R

$$\prod_{t=0}^{T}\prod_a p(a)^{Y_{a,t}}\left(1-p(a)\right)^{N_{a,t}}$$

- Given the user activity log, we can compute α to maximize the above likelihood
- Compute α and β that maximize $\prod_a p(a)^{Y_a}\left(1-p(a)\right)^{N_a}$ where $Y_a = \sum_t Y_{a,t}$ and $N_a = \sum_t N_{a,t}$
- α measures the correlation between nodes.

Shuffle test - Influence vs. Correlation

In studies modeling influence, influence is determined by timestamps

Key idea: if influence does not play a role, the timing of activation should be independent of the timing of other actors / nodes

➔ **even if we randomly shuffle the timestamps of user activities, we should obtain similar social correlation** (coefficient values)

Test for Influence:

Shuffle the timestamps of user activities. If the new estimate of social correlation is significantly different from the estimate based on the user activity log, then there is evidence of influence.

Shuffle test - Influence vs. Correlation

G a social network, $W = \{w_1, \dots, w_\ell\}$ the set of nodes activated during whole measuring with T time steps; node w_i was activated at time t_i

1. For all $t = 1, \dots, T$: compute $Y_{a,t}$ and $N_{a,t}$.
2. Compute α and β that maximize $\prod_a p(a)^{Y_a}\big(1 - p(a)\big)^{N_a}$ where $Y_a = \sum_t Y_{a,t}$ and $N_a = \sum_t N_{a,t}$ and $p(a) = \frac{e^{\alpha \ln(a+1)+\beta}}{1+e^{\alpha \ln(a+1)+\beta}}$.
3. Pick a random permutation π of $\{1, \dots, \ell\}$. Set the activation time of node w_i to $t_i' = t_{\pi(i)}$. Based on $t_1', \dots, t_\ell'$, compute $Y_{a,t}'$ and $N_{a,t}'$.
4. Again, compute α' and β' that maximize $\prod_a p'(a)^{Y_a'}\,\big(1 - p'(a)\big)^{N_a'}$ where $Y_a' = \sum_t Y_{a,t}'$ and $N_a' = \sum_t N_{a,t}'$ and $p'(a) = \frac{e^{\alpha' \ln(a+1)+\beta'}}{1+e^{\alpha' \ln(a+1)+\beta'}}$.
5. The model exhibits no social influence if the values of α and α' are close to each other.

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Homophily, Social Influence, and Affiliation

- Homophily Test
- Selection and Social Influence
- Affiliation Networks
- Schelling Model of Segregation
- Positive and Negative Relationships